Q-Reliability Lab

Computational-basis success can remain perfect while phase-flip noise destroys the target Bell-state relation

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Independent research project

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Classical density-matrix simulation with deterministic finite-shot sampling; no quantum-hardware data.

Outline

1. Motivation and precise research question
2. Phase-flip model and analytic predictions
3. Basis-sensitive success metrics
4. Exact density-matrix sweep
5. Finite-shot experiment and uncertainty
6. Channel–metric sensitivity comparison
7. Reproducibility, limits, and implications
8. Conclusions and future work

Seminar claim

For a Bell-state benchmark, a success definition based only on accepted computational-basis populations cannot certify the relative phase of the state.

Source: Project source, analytic record, and committed experiment artifacts.

Question and model

A Bell-state benchmark under a first-qubit phase-flip channel

Motivation: accepted outcomes can hide state damage

- Bell-state preparation targets $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$.
- A computational-basis task can accept both 00 and 11, reporting $S_Z = P(00) + P(11)$.
- $|\Phi^+\rangle$ and $|\Phi^-\rangle = (|00\rangle - |11\rangle)/\sqrt{2}$ have the same accepted populations.
- Their relative phase differs, so agreement with the intended target differs.

Concrete failure mode

At $p = 1$, the output becomes $|\Phi^-\rangle$:

$$S_Z = 1 \qquad F_{\Phi^+} = 0.$$

Reliability question

Does a perfect task success score actually establish preparation of the desired Bell state?

Source: Analytic setup in `paper/main.tex`, Introduction and Methodology.

Bell state and phase-flip channel

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}},$$

$$\rho_+ = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

$$E_0 = \sqrt{1-p}\, I_4, \qquad E_1 = \sqrt{p}\,(Z \otimes I).$$

- Ordered basis: $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$.
- Noise acts only after the ideal Bell circuit.
- $p \in [0, 1]$ denotes the channel probability.

Kraus completeness

$$\sum_k E_k^\dagger E_k = (1-p) I_4 + p(Z \otimes I)^2 = I_4.$$

Physical interpretation

$(Z \otimes I)\,|\Phi^+\rangle = |\Phi^-\rangle$, so the channel mixes two orthogonal Bell states.

Source: Kraus construction and completeness: src/phase_error_blind_spot.py.

Central question and hypothesis

Question

For the stated phase-flip model, which measured quantities remain informative as p changes from zero to one?

Hypothesis

The computational-basis correlated-success metric is structurally insensitive to the phase flip, while fidelity, signed coherence, and an X-basis correlation expose the change.

- Exact sweep: 101 values, $p = 0.00, 0.01, \ldots, 1.00$.
- Finite-shot sweep: five p values, three shot budgets, two bases, 100 repetitions.

Claim boundary

The study reproduces a known measurement-basis limitation as a transparent benchmark. It does not introduce a new noise model or estimate a hardware error rate.

Falsifiable contrast

If S_Z falls with p under this model, the analytic derivation or implementation is incorrect.

Source: Scope statement: `docs/self_audit.md`; model: `paper/main.tex`.

Exact analytic baseline

$$\rho(p) = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1-2p \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1-2p & 0 & 0 & 1 \end{pmatrix}.$$

$$S_Z(p) = 1, \qquad F_{\Phi^+}(p) = 1 - p,$$

$$\mathrm{Re}\,\rho_{00,11}(p) = \frac{1-2p}{2}, \qquad |\rho_{00,11}(p)| = \frac{|1-2p|}{2}.$$

p	S_Z	F	$\mathrm{Re}\,\rho_{00,11}$
0.00	1.00	1.00	0.50
0.50	1.00	0.50	0.00
1.00	1.00	0.00	-0.50

Endpoint interpretation

$p = 0.5$ removes coherence. $p = 1$ reverses its sign and produces a state orthogonal to the target.

Source: Values from `experiments/phase_error_sweep.csv`; formulas in `paper/main.tex`.

Metrics and exact sweep

Success definitions determine which errors a benchmark can observe

Z-basis and X-basis correlated success

$$
\begin{aligned}
S_Z(\rho) &= \rho_{00,00} + \rho_{11,11}, \\
\Pi_X &= |{+}{+}\rangle\langle{+}{+}| + |{-}{-}\rangle\langle{-}{-}|, \\
S_X(\rho) &= \mathrm{Tr}(\Pi_X \rho) \\
&= \tfrac{1}{2}(\rho_{00,00} + \rho_{11,11} + \rho_{00,11} + \rho_{11,00}).
\end{aligned}
$$

- Code evaluates S_X from the projector, rather than hardcoding $1 - p$.
- S_Z uses only populations; S_X also uses off-diagonal coherence.

Metric	Information tested
S_Z	Computational-basis correlated populations
S_X	Relative phase through X-basis correlation
Fidelity	Agreement with the specified $\|\Phi^+\rangle$ target
Signed coherence	Direction and size of $\rho_{00,11}$

Source: Projector metric: `src/phase_error_blind_spot.py`; tests: `tests/test_x_basis_diagnostic.py`.

Structural blindness of computational-basis success

$$\rho_{00,00}(p) = \tfrac{1}{2},$$
$$\rho_{11,11}(p) = \tfrac{1}{2},$$
$$S_Z(p) = \rho_{00,00}(p) + \rho_{11,11}(p) = 1.$$

- The diagonal populations remain fixed under a Z operation on the first qubit.
- Only the cross terms acquire the factor $1 - 2p$.
- Therefore additional Z-basis samples cannot reveal p in this model.

Structural, not numerical

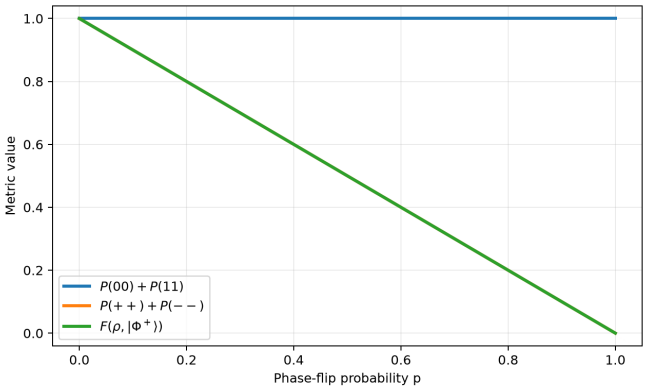
The constancy follows from the channel and measurement projector. It does not result from a coarse probability grid, floating-point rounding, or insufficient shot count.

Benchmark implication

A task score may measure a valid but incomplete property of the output state.

Source: Index-level derivation: `paper/main.tex`, Eq. $\rho(p)$ and metric definitions.

Exact density-matrix sweep



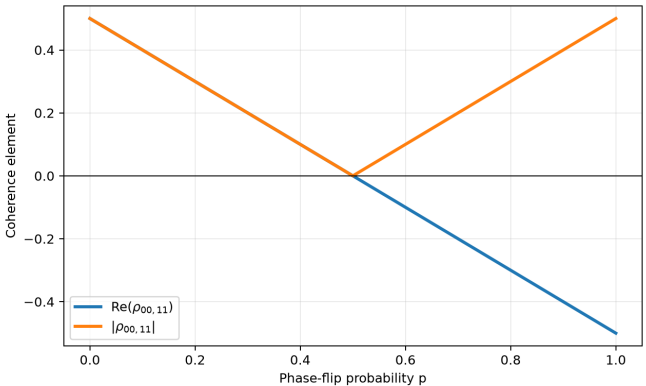
- The exact simulator applies the Kraus map to ρ_+ at 101 values of p.
- S_Z remains equal to one across the entire sweep.
- S_X and fidelity both decrease linearly as $1 - p$.
- The visual agreement reflects shared analytic predictions, checked by parameterized tests.

Conclusion

The selected Z-basis task accepts an output whose target-state fidelity falls continuously from one to zero.

Source: Generated by `python -m src.phase_error_blind_spot`; figure: `paper/figures/success_vs_fidelity.png`.

Signed coherence distinguishes loss from phase reversal



- $|\rho_{00,11}|$ decreases to zero at $p = 0.5$ and returns to 0.5 at $p = 1$.
- The magnitude alone cannot distinguish $|\Phi^+\rangle$ from $|\Phi^-\rangle$.
- $\mathrm{Re}\,\rho_{00,11}$ changes sign and records the phase reversal.

Interpretation

At $p = 1$, coherence survives but has the wrong sign for the target state.

Source: Generated by `python -m src.phase_error_blind_spot`; data: `experiments/phase_error_sweep.csv`.

Finite-shot evidence

Sampling uncertainty changes precision, not metric sensitivity

Finite-shot experiment design

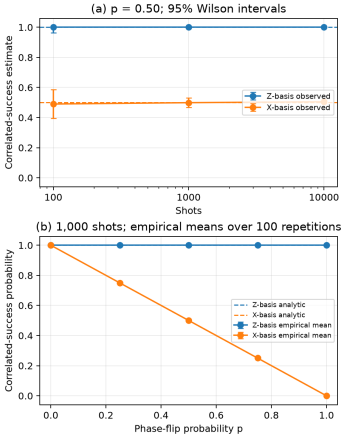
Design field	Committed specification
p values	$0, 0.25, 0.50, 0.75, 1.00$
Shot budgets	$100, 1{,}000, 10{,}000$
Measurement bases	Z and X
Repetitions	100 per condition
Root seed	20261030
CSV rows	3,000
Interval	two-sided 95% Wilson

$$K_Z = N,$$
$$K_X \sim \text{Binomial}(N, 1-p),$$
$$\widehat{S}_B = K_B/N.$$

- Each row records its child seed, success count, estimate, analytic value, and interval endpoints.
- These observations are simulated binomial samples, not hardware measurements.

Source: Design and schema: `src/finite_shot_phase_error.py`; data: `experiments/finite_shot_sweep.csv`.

Finite-shot results at $p = 0.50$



- Every Z-basis observation reports correlated success one, for every shot budget.
- X-basis estimates remain close to the analytic value $S_X = 0.5$.
- The X-basis interval narrows as N increases.
- More samples improve an informative estimator; they do not repair an uninformative measurement basis.

Conclusion

Finite sampling introduces uncertainty around S_X but leaves the Z-basis blind spot unchanged.

Source: Generated by python -m src.finite_shot_figures; figure data from experiments/finite_shot_sweep.csv.

Wilson intervals quantify sampling precision

N	$\widehat{S}_X$	95% Wilson lower	upper	
100	0.4900	0.3942	0.5865	All rows
1,000	0.4990	0.4681	0.5299	
10,000	0.5053	0.4955	0.5151	

use $p = 0.50$, basis X, repetition 0. The analytic success probability equals 0.5000.

- Wilson intervals remain bounded at all-success and zero-success endpoints.
- Their width reflects finite sample size, not uncertainty about the model parameters.
- A Wilson interval is a frequentist confidence interval, not a posterior probability.

Source: Exact displayed values from `experiments/finite_shot_sweep.csv`; interval implementation in `src/finite_shot_phase_error.py`.

Comparative benchmark

A metric can be blind to one Pauli error while detecting another

Noise-channel and metric sensitivity matrix



- Cells report the maximum absolute metric change over $p \in [0, 1]$.
- S_Z is structurally blind to phase flips.
- S_X is structurally blind to bit flips.
- Every listed metric changes under the specified depolarizing channel.

Conclusion

No single listed metric detects every modeled Pauli noise type; benchmark design must match the error mechanism.

Source: Generated by `python -m src.channel_metric_matrix`; data: `experiments/channel_metric_matrix.csv`.

Channel comparison at the sweep endpoints

Channel at $p = 1$	S_Z	S_X	F	$\mathrm{Re}\,\rho$	$\lvert\rho\rvert$
Bit flip	0.00	1.00	0.00	0.00	0.00
Phase flip	1.00	0.00	0.00	-0.50	0.50
Depolarizing	0.50	0.50	0.25	0.00	0.00

ρ denotes the target-basis coherence element $\rho_{00,11}$.

- The bit-flip and phase-flip cases exchange which basis metric remains constant.
- Depolarizing noise uses weights $1 - 3p/4, p/4, p/4, p/4$.
- Its target-fidelity endpoint equals $1/4$, consistent with the maximally mixed state.

Source: Values from `experiments/channel_metric_matrix.csv`; channel formulas in `src/channel_metric_matrix.py`.

Reproducibility and interpretation

Evidence, validation boundary, and claims supported by the model

Reproducibility and engineering evidence

Evidence	Project mechanism
Exact analytics	Parameterized regression tests at five fixed p values
Physical channel checks	Trace, Hermiticity, PSD, Kraus completeness, invalid-input tests
Data regeneration	Deterministic CSV sweeps and committed generated table
Finite-shot repeatability	Fixed root seed and stored child seed per row
Automation	GitHub Actions runs tests, regeneration checks, paper build, Ruff, and CodeQL

Source: Workflows: `.github/workflows`; tests: `tests/`; build notes: `docs/reproducibility.md`.

- Dependencies for CI are pinned in `requirements-ci.txt`.
- The CI check rejects uncommitted changes to deterministic data artifacts.
- The visual lab implements the same phase-flip metrics and is tested against the Python model.

Reproducibility boundary

These checks validate the stated simulation and artifacts; they do not validate quantum hardware behavior.

Threats to validity and scope limitations

Model limitations

- Classical density-matrix and binomial simulation only.
- One target state, one post-circuit noise location, and three Pauli channels.
- Ideal preparation and measurement in the exact model.

Omitted hardware effects

SPAM error, readout error, crosstalk, calibration drift, coherent control error, and device-specific CNOT noise.

What the study establishes

Within the declared channel and measurement model, the selected Z-basis success metric cannot identify phase-flip strength.

What it does not establish

The study does not measure an IBM or other device error rate, demonstrate a fault-tolerant protocol, or provide a complete process characterization.

Source: Detailed audit and claim boundaries: `docs/self_audit.md` and `paper/main.tex`, Discussion.

Discussion: implications for benchmark design

- A metric can remain numerically stable because its measurement projector excludes the affected degree of freedom.
- More shots reduce uncertainty only for variation that the measurement can observe.
- Fidelity detects mismatch with a chosen target but does not fully characterize a general quantum process.
- Signed coherence adds directional phase information that coherence magnitude discards.

Practical reading of a benchmark score

Before interpreting a high score as reliable state preparation, specify: the target state, accepted outcomes, measurement basis, noise model, and uncertainty model.

Design implication

Use complementary observables when the task specification depends on both population and phase information.

Source: Interpretation grounded in the exact and finite-shot results shown in this deck.

Conclusion

1. The first-qubit phase-flip channel preserves $P(00) + P(11)$ for the Bell-state benchmark, so $S_Z(p) = 1$ across the full sweep.
2. The target fidelity and X-basis correlated success both equal $1 - p$; they expose degradation missed by the computational-basis score.
3. At $p = 1$, the output is the coherent but target-orthogonal state $|\Phi^-\rangle$, yielding $S_Z = 1$ and $F_{\Phi^+} = 0$.
4. Deterministic data, finite-shot intervals, tests, CI, and an interactive visual lab make the benchmark inspectable and reproducible.

Honest contribution

A reproducible quantitative demonstration of a known metric blind spot, framed as a compact reliability benchmark rather than a novel physical discovery.

Source: All conclusions are limited to the declared classical simulation model.

Future work

- Extend the state family beyond $|\Phi^+\rangle$ and vary the noise location within an explicit circuit.
- Compare coherent over-rotation, amplitude damping, readout error, and correlated multi-qubit noise.
- Add tomography-inspired or randomized-benchmarking-style observables where their assumptions are explicit.
- Run a separately calibrated hardware experiment with documented device, backend date, transpilation, and shot protocol.
- Report hardware uncertainty separately from simulation uncertainty.
- Evaluate whether metric blind spots persist under realistic task-level acceptance rules.

Next empirical step

Hardware data would require a new experiment design; the present simulation supplies analytic controls and regression baselines for that extension.

Source: Proposed extensions, not results of the current study.

References



M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press, 10th Anniversary Edition, 2010. doi:10.1017/CBO9780511976667.



J. Preskill. Quantum Computing in the NISQ Era and Beyond. *Quantum*, 2:79, 2018. doi:10.22331/q-2018-08-06-79.



C. J. Wood, J. D. Biamonte, and D. G. Cory. Tensor Networks and Graphical Calculus for Open Quantum Systems. *Quantum Information and Computation*, 15(9–10):759–811, 2015.



Z.-J. Liu. *Q-Reliability Lab*. Version 0.1.0, 2026. Repository: `github.com/Raina-ya/q-reliability-lab`.

Source: Bibliographic entries match `paper/references.bib` and `CITATION.cff`.

Backup: output-state derivation

$$\begin{aligned}\mathcal{E}_p(\rho_+) &= (1-p)\rho_+ + p(Z\otimes I)\rho_+(Z\otimes I)\\ &= (1-p)\,|\Phi^+\rangle\langle\Phi^+| + p\,|\Phi^-\rangle\langle\Phi^-|\,.\end{aligned}$$

$$\rho(p) = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1-2p \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1-2p & 0 & 0 & 1 \end{pmatrix}.$$

- Diagonal entries remain $1/2$.
- The coherence crosses zero at $p = 1/2$.
- The endpoint at $p = 1$ has negative coherence, not zero coherence.

Source: Full derivation: `notes/2026-10-29-analytic-phase-flip-record.md` and `paper/main.tex`.

Backup: representative exact sweep values

p	S_Z	S_X	F	$\mathrm{Re}\,\rho_{00,11}$
0.00	1.00	1.00	1.00	0.50
0.25	1.00	0.75	0.75	0.25
0.50	1.00	0.50	0.50	0.00
0.75	1.00	0.25	0.25	-0.25
1.00	1.00	0.00	0.00	-0.50

Cross-check

The table equals the committed CSV values and the analytic relations used in the regression tests.

Source: `experiments/phase_error_sweep.csv` and `paper/generated/phase_error_results_table.tex`.

Backup: deterministic regeneration commands

Exact sweep, figures, and paper table

```
python -m src.phase_error_blind_spot --output experiments/phase_error_sweep.csv
--figures paper/figures --table paper/generated/phase_error_results_table.tex
```

Finite-shot data and figure

```
python -m src.finite_shot_phase_error --output experiments/finite_shot_sweep.csv
--root-seed 20261030
```

```
python -m src.finite_shot_figures --input experiments/finite_shot_sweep.csv --output
paper/figures/finite_shot_detectability.png
```

Channel matrix

```
python -m src.channel_metric_matrix --output experiments/channel_metric_matrix.csv
--figure paper/figures/channel_metric_sensitivity_matrix.png
```

Source: CI verifies the deterministic CSV and generated table artifacts.